

ESP32-WROOM-32U, shared USB-C charging/programming, protected 1S power, 3.3 V buck-boost and MAX98357A



Place C12/C13 and the EN RC parts close to U1. Keep the U.FL connector accessible at the board edge

Speaker output is bridge-tied: neither speaker wire may connect to GND.

- Use 4 layers if possible: signal / solid GND / power / signal.
- Keep U.F.L enclosure-accessible and keep antenna coax away from the Class-D output.
- Route USB D+/D- as a short 90-ohm differential pair; keep both CC resistors.
- Use wide copper for VSYS, battery and speaker; stitch exposed pads to GND.
- TP4056 is a 4.2 V charger; do not use a 4.35 V-only cell profile.
- Recheck OLED and INMP441 availability before ordering Standard PCBs.

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**Title: WalkieTalkie Main PCBA**

Size: A3

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